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absstem INDUSTRIAL

Quotation No : Q24000225

Date : 22-Apr-2024

JP Minda Group
GP-14, HSIIDC Industrial Estate, Sector 18
GURGAON
HARYANA
122001

Kind Attention : Mr./Ms. Kameshwar Singh

Dear Sir,

We understand that in industry, reliability is not just a requirement,it is the foundation of success.
That is why Absstem Industrial PSA Generator is more than just a product; it is a commitment to excellence, efficiency, and uninterrupted performance.

Every component is designed with precision and innovation, ensuring that each molecule of oxygen and nitrogen is delivered with the utmost purity.

With fully automated operation, our system seamlessly integrates into your processes, reducing manual intervention and maximizing productivity.

Moreover, our industry-leading air ratio ensures that you achieve the highest efficiency, optimizing energy consumption while maintaining peak performance.
And because we believe in standing by our craftsmanship, we offer an unmatched-10 year warranty on sieves, giving you the peace of mind that your investment is protected for the long run.

We have attached the detailed quotation for your review.

Should you have any questions or require further customization to meet your specific needs, please do not hesitate to reach out.

We are here to provide not just a product but a solution that breathes life into your operations.

We look forward to the opportunity to serve you and build a lasting partnership.

Thanks and Regards,

For Absstem Industrial Solutions Pvt. Ltd.

Arifa Zaidi

Assistant Manager Business Development

9218108184

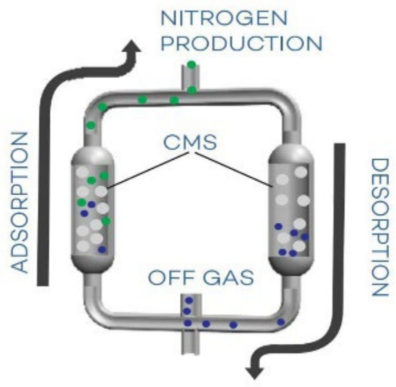
www.absstem.com

nGEN POWERED BY ABSSTEM INDUSTRIAL

Industrial
PSA Nitrogen Generator

absstem
nGEN
absstem
PSA
Nitrogen Generator

PSA Technology



PSA (PRESSURE SWING ADSORPTION)

Industrial Nitrogen Generator mainly consists of two adsorbers filled with carbon molecular sieves. Pressurized air is passed through one of these adsorbers, carbon having the property to attract oxygen, adsorbs oxygen from air and Nitrogen is allowed to pass through and move into the storage tank as the final product. As the adsorber approaches saturation, the air is allowed to pass through the other adsorber while this adsorber is being regenerated to ensure a continuous supply of Nitrogen. This changeover takes place through a specially designed microcontroller to ensure a hassle-free operation. The quality of Nitrogen produced is monitored by a Nitrogen analyser.

Key Features

- Shipped ready for installation
- 5 minutes to run and never have to stop it!
- Easy installation
- Low operating expenses compared to gas cylinders/liquid Nitrogen
- Maintenance costs are very less
- Embedded Safety Interlocks
- Space-saving design
- Eliminates all hazards which are there with gas cylinder or liquid Nitrogen
- Longevity: at least 10 years

Description

a.	Product Name	Nitrogen Gas Generator
b.	Model Range	**NGEN**
	Technology	PSA (Pressure Swing Adsorption)

Specifications

a.	Nitrogen Flow Rate	As per Performance Sheet
b.	Nitrogen Purity	As per Performance Sheet
c.	Nitrogen Delivery Pressure	5.5 Barg (Higher Pressure Options Available)
d.	Ambient conditions	
	Temperature	0 to 50 Deg. Celsius
	Relative Humidity	up to 80 % (Max.)
	Altitude	up to 1000 m above Sea level

Scheme Offered – (PSA Only)

This scheme will comprise of

- Air package with inlet air filtration system
  (Air compressor, Air dryer, Water Separator, Oil filters)
- Industrial Nitrogen generator
- Storage Tank
- Nitrogen Analyzer
- Alarm panel

Unique Features of Absstem

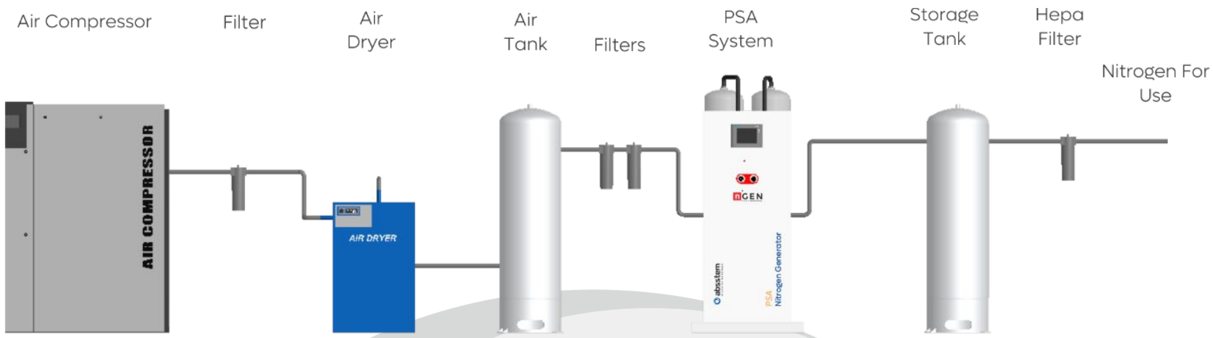
a. All the fabricated items are cleaned and degreased following our strict quality policy.
b. All the Instruments are specially manufactured to make them suitable for Nitrogen gas.
c. Guaranteed 10+ years of healthy life.
d. We provide the purest **Nitrogen** in Industry.
e. Data logging and Graphical Reports for all parameters available.
f. Test certificates of all instruments are available on demand.

Onsite Gas Generation vs Liquid Tank/Cylinder Alternative

Non-stop availability: 24 hours a day, 7 days a week
Easy integration within existing installations
Your own independent supply of industrial gas
Significant economies for scale and lower operational costs: no rental charges, logistical expenses and bulk user evaporation losses
No safety hazards when handling high-pressure cylinders

Equipment Description

The Industrial Nitrogen Generator consists of


Air Compressor
Filter
Air Dryer
Air Tank
Filters
PSA System
Storage Tank
Hepa Filter
Nitrogen For Use
AIR COMPRESSOR
AIR DRYER
nGEN
absstem
PSA Nitrogen Generator

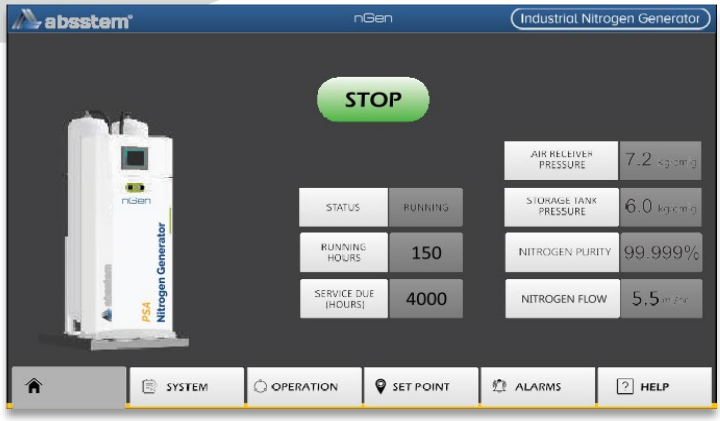
Air Compressor	An Air compressor is used to compress the atmospheric air to a pressure required for the separation of **Nitrogen** from the air.
Refrigeration dryer	Compressed air then passes through a refrigerated dryer to remove moisture (which is harmful to sieves) from the air.
Air Tank	Dry air is collected in an air tank to remove the pulsations which serve the dynamic (changing) requirement of air by the PSA system.
Multistage Filtration system	Before air enters the PSA system, it is passed through multistage filtration removing moisture, particulates and oil content to the specified levels.
PSA system	Pressurized air is passed through PSA Towers filled with Molecular sieves to separate **Nitrogen** from Air.
Storage Tank	Industrial Grade **Nitrogen** produced from NGEN is stored in this tank to supply a continuous flow of **Nitrogen**.
Nitrogen Analyzer	Nitrogen produced is monitored by a **Nitrogen** analyser. In case, O2 purity goes below the specified level, it gives an alarm and switches it to vent mode.

HD Touch Screen Display

(User-friendly touchscreen interface for easy operation and maintenance)

Absstem industrial Nitrogen generator complies with all industry standards and regulation

- Alarm management system
- Plug and play
- Fully automatic
- Process data display
- Consumables running hours display
- Data logging via USB drive
- Graphical representation


absstem
nGen
Industrial Nitrogen Generator
STOP
nGen
PSA Nitrogen Generator
SYSTEM
OPERATION
SET POINT
ALARMS
HELP

STATUS RUNNING
RUNNING HOURS 150
SERVICE DUE (HOURS) 4000
AIR RECEIVER PRESSURE 7.2 kg/cm²g
STORAGE TANK PRESSURE 6.0 kg/cm²g
NITROGEN PURITY 99.999%
NITROGEN FLOW 5.5 m³/hr

Additional upgrades

- Digital flow meter with totaliser
- High-pressure Nitrogen tank
- Cylinder manifold and alarm management system
- Remote monitoring

Plant Operation

- Unit operates 24-hrs a day, 7-days a week reliably, automatically & safely

Automation

- The Plant is fully automatic and needs only electrical power for running the Air compressor
- The machine automatically enters into standby mode when the level in the Storage Tank reaches its set limit. This helps to increase the life of the Air compressor.

International Quality standards

A strict quality procedure, following the leading international pharmacopoeia and Industrial directives, is followed in the selection of all mechanical and electrical instruments, vessel fabrication and designing of Industrial Nitrogen generators.

TUV NORD TÜV NORD CERT INDIA ISO 14001

TUV NORD TÜV NORD CERT INDIA ISO 13485

RIR CERTIFICATION Partners in Growth ISO 45001

TUV NORD TÜV NORD CERT INDIA ISO 9001

Itemised Supply Description

No.	Item description	Make	Qty
a	Air Compressor with Filters	ABSSTEM APPROVED	Quoted Separately
b	Refrigeration Dryer	ABSSTEM APPROVED	Quoted Separately
c	Air Tank	ABSSTEM	Quoted Separately
d	Set of Filters	ABSSTEM	1 no.
e	PSA unit (NGEN)	ABSSTEM	1 no.
f	Electrical Power Panel with PLC and all cabling	ABSSTEM & MITSUBISHI	1 no.
f	Digital Nitrogen Analyzer	ABSSTEM	1 no.
g	Storage Tank	ABSSTEM	Quoted Separately
h	Outlet product gas Filter	ABSSTEM	1 no.

Electrical Protection

The client should make proper arrangements for earthing/grounding systems for the Industrial **Nitrogen** Generator at their facility.

Enclosure Panel

- Ergonomically designed metal enclosure with powder-coated finish for longer life.
- High-strength IP-rated panel fittings from Germany for a robust enclosure.
- Fan filters from **Rittal** are IP-55 rated making the electrical enclosure's internal environment dust-free.
- Easy connectivity to any place of installation with req. port connections.
- High-strength base frame designed for higher load and easy transportation.
- Vessels are designed per ASME section VIII DIV I. Vessels are epoxy paint coated which is considered good for longevity and corrosion resistance.
- All electrical items are CE certified with operating voltage at 24 VDC

Annual Maintenance Contract (AMC)

Our in-house support Absstem shield is 24*7 open and we assure a timely service maintenance schedule. Per the schedule mentioned in the operational manual, we suggest the operator change the filters by following a simple protocol.

It is very easy to change or replace as our commissioning engineer plus our commissioning engineer and tech support office will always guide you about the activity.

Best ROI Guaranteed

ROI within 6 to 18 months when compared to conventional methods

Generator performance depends on the component selection. Our processes make this job easier as all equipment is selected based on more than 3 decades of research. NGEN is a fully automatic (operator-free operation) that ensures peace of mind.

We deliver what we commit!!

Thank You

Product Description

Product	Description	Make	Qty
nGen Nitrogen Generator	nGEN 70 P Nitrogen		1

Caption	Specification
Gas Generated	Nitrogen
Gas Pressure	upto 6.5 Barg(Higher options Available)
Flow @ 99.99 %	68 Nm3/hr
Flow @ 99.995 %	51 Nm3/hr
Flow @ 99.999 %	39 Nm3/hr
Flow @ 99.9995 %	31 Nm3/hr
Compressed Air Requirement	249 CFM
Module	High Purity PSA system Only
Air Compressor And Drier	In your Scope / Quoted Separately

Product	Description	Make	Qty
nGen Gas Storage	nGEN 55 / 60 / 65 / 70 ST Nitrogen		1

Caption	Specification
Water Volume	6000 Liters
Gas Stored	Nitrogen
Gas Stored Volume	39 M3
Working Pressure	6.5 Barg
Hydrotest Pressure	9 Barg

Commercial Offer

S No	Item Code	Item Name	Description	HSN	Qty	Unit Price	Discount	GST(%)	Amount
1	nGEN 70 P	nGEN 70 P	nGen Nitrogen Generator	840510	1	22,43,000.00	4,12,936.30	18.00	18,30,063.70
2	nGEN 60 ST	nGEN 55 / 60 / 65 / 70 ST	nGen Gas Storage	840510	1	4,00,000.00	80,000.00	18.00	3,20,000.00
								Sub Total	21,50,063.70

GST(%) @18.00	3,87,011.47
Grand Total ()	25,37,075.17

Terms And Conditions

1.	SUPERVISION OF ERECTION & COMMISSIONING	There shall be no commissioning charges, however you shall arrange freeTo & Fro Air Ticket, lodging / boarding, and local conveyance for our Engineers during supervision of Installation of Gas Plant at site on free of cost basis. Our Engineers will be there for Supervision only. Any manpower / fitter required for installation work shall be provided by you.
2.	P&F	3% Extra.
3.	Freight	Extra at Actuals.
4.	Insurance	In your Scope.
5.	Payment Terms	50% Advance with your Prestigious Purchase Order, 40% + GST against PI before dispatch and Balance 10% on commissioning.
6.	Delivery	4 Weeks from the PO and Advance. Rest can be confirmed for Availability.
7.	Warranty	We hereby confirm that, our equipment shall have Supplier's warranty for a period of 12 months from the date of Commissioning or 18 months from date of Dispatch whichever is earlier.
8.	GST	As per the Commercial Table.
9.	Remarks	.
10.	Absstem GST Details	06AAWCA9492P1ZF - ABSSTEM INDUSTRIAL SOLUTIONS PRIVATE LIMITED

Hope you will find our offer in line with your requirement.

Should you require any further clarification/information, please feel free to contact us.

Thanking you, and assuring you of our best attention at all times.

Very Yours truly,
For Absstem Industrial Solutions Pvt. Ltd.
Arifa Zaidi
Assistant Manager Business Development
9218108184
www.absstem.com